

# EM\_MLE\_Derivation

This document will detail the derivations required for the MLEs of all of the tensor variate distributions.

## Univariate Generalized Hyperbolic

First, consider the simpler case where  $X$  is generated from a generalized hyperbolic distribution. Under the Barndorff-Nielsen construction, we can create these draws using

$$X = \mu + W\alpha + \sqrt{W}Z$$

where  $\mu$  is a vector,  $\alpha$  is a vector,  $Z \sim N(0, \Sigma)$ . The distribution of  $W$  is what determines the distribution of  $X$ . In this case, let  $W$  be from the generalized inverse Gaussian distribution denoted as  $W \sim \text{GIG}(a, b, \lambda)$ .

Thus, the density function of  $W$  is

$$f(w|a, b, \lambda) = \frac{(\frac{a}{b})^{\lambda/2} w^{\lambda-1}}{2K_{\lambda}(\sqrt{ab})} \exp \left\{ \frac{-1}{2} \left( aw + \frac{b}{w} \right) \right\}$$

where

$$K_{\lambda}(x) = \frac{1}{2} \int_0^{\infty} y^{\lambda-1} \exp \left\{ \frac{-x}{2} \left( y + \frac{1}{y} \right) \right\} dy$$

is the modified Bessel function of the third kind.

We are interested in computing the MLEs of this function given draws. Thus, we want to rewrite this data in terms of only the random draws  $x$ .

Consider again the construction used to create  $X$ . We can easily define the distribution of  $f(X|w)$ . If we knew the underlying latent draw  $w$ , then  $f(X|w) \sim N(\mu + w\alpha, w\Sigma)$ . We also know the distribution of  $W$ .

$$\begin{aligned}
f(x) &= \int f(x|w)f(w)dw \\
&= \int (2\pi)^{-k/2} \det(w\Sigma)^{-1/2} \exp\left(\frac{-1}{2}(x - \mu - w\alpha)^T w\Sigma^{-1}(x - \mu - w\alpha)\right) dw
\end{aligned}$$

## Tensor Variate Skewed Distribution

In the tensor variate case, the normal variance mean mixture model is an efficient way to introduce skewness. A random tensor  $\mathcal{X}$  can be written as

$$\mathcal{X}_i = \mathcal{M} + W_i \mathcal{A} + \sqrt{W_i} \mathcal{V}_i$$

where  $\mathcal{M}$  is a location tensor,  $\mathcal{A}$  is a skewness tensor,  $\mathcal{V} \sim \mathcal{N}_{n^*}(\mathbb{0}, \bigotimes_{d=1}^D \Sigma_d)$  and  $W > 0$  is a positive random variable. The distribution of  $\mathcal{X}$  will depend on the distribution of  $W$ .

We are interested in obtaining the MLEs of the distribution  $X$ .

If we knew the latent values  $W_i$ , then the distribution of  $X_i$  would be normal.

$$X_i | W_i = w_i \sim N_p(W_i \mathcal{A}, W_i \bigotimes_{i=1}^k \Sigma_k)$$

Let the distribution of  $w$  be represented as  $h(w)$ . Then the joint density can be written as

$$f(X_i, W_i; \theta) = f(X_i | w_i) f(w_i).$$

Thus, we can represent the joint distribution as a product of the tensor variate normal PDF and the PDF of  $W$ .